Magnetism: electrostatics plus relativity

Nonrelativistic test charges around steady currents; c = 1/sqrt(mu0 eps0).

Two charges in motion

$$F_m = \frac{\mu_0}{4\pi}\frac{q_1 q_2 v_1 v_2}{r^2}$$

$$\frac{F_m}{F_e} = \frac{v_1 v_2}{c^2}$$

Neutral wires hide the electric force but not the magnetic one. Between two free charges moving parallel, the Biot-Savart field of one crossed with the velocity of the other gives F_m = (mu0/4pi) q1 q2 v1 v2 / r², while the Coulomb force is (1/4pi eps0) q1 q2 / r². Using mu0 eps0 = 1/c² the ratio is v1 v2 / c².

WHY / The tiny ratio is the reason magnetism looks like a separate force at laboratory speeds; it is the first relativistic correction to electrostatics.

WATCH / Only for v much less than c and parallel motion does this one-line ratio hold; perpendicular compone

Charge density changes with frame

$$\lambda' = -\gamma I v/c^2, \qquad F = qvB$$

In the wire frame the positive and negative line densities cancel. A test charge moving at v sees length-contraction differences and a net force that, expanded to first order in v/c, equals q v B with B = mu0 I / 2 pi r. The constant mu0 = 1/(eps0 c²) emerges, not postulates.

WHY / Deriving magnetism from one boosted frame explains why the magnetic constant is tied to c: it is not an independent phenomenon.

WATCH / The full Lorentz transformation of fields is the proper language; the force-balance sketch is the exam-level derivation, not a replacement for Maxwell theory.

CHECK / Length contraction changes the charge density of a neutral wire in the moving frame; the deficit of ions is exactly the current.

The worked comparison

$$F_e = ke^2/r^2 \gg F_m \text{ at lab speeds}$$

$$B = \frac{1}{c^2} v \times E$$

Two electrons at 10^6 m/s one centimetre apart: F_e about 2.3e-24 N and F_m smaller by 1e-11. In wires the electric force cancels to neutrality and the magnetic residue is what remains measurable. A moving charge field is B = v cross E over c² exactly in the nonrelativistic limit.

WHY / The estimator F_m/F_e = v1 v2/c² is a one-line validity check on any claimed magnetic dominance.

WATCH / For charges near light speed the magnetic force is comparable to the electric one; relativity is then not a correction but the description.

CHECK / The worked comparison: two protons at 10^6 m/s, 1 cm apart: Fe = 2.3e-24 N repulsive, Fm = 2.6e-29 N attractive.

Limits of the classical picture

$$g = 2 \text{ (no classical account)}$$

$$\mu_0 \varepsilon_0 = 1/c^2$$

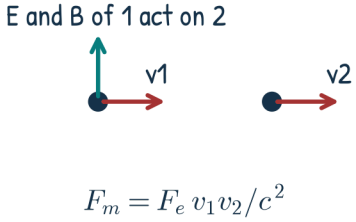
Spin, the g factor, exchange coupling and superconductivity have no classical derivation; the relativistic picture covers orbital moments and forces between currents. Radiation from accelerating charges lies in the electromagnetic-waves module.

WHY / Stating the boundary of the classical model is part of the physics; the constant 1/c² is the bridge, not the whole road.

WATCH / Do not present v cross E / c² as valid at relativistic speed; transform the fields properly there.

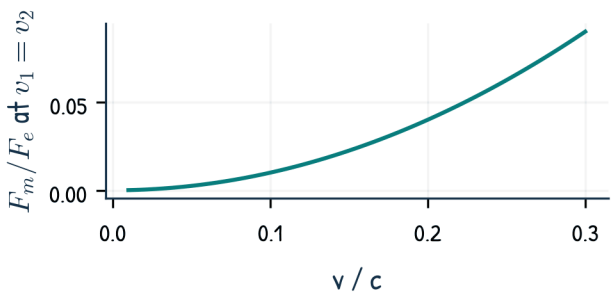
CHECK / Classical magnetostatics assumes steady currents and non-relativistic test charges; near c the full Lorentz transformation is required.

Two moving charges: the magnetic term is the relativistic residue



$$F_m = F_e\, v_1 v_2/c^2$$

Magnetic force is smaller than electric by v squared over c squared



REMEMBER / The magnetic force between parallel moving charges is the electric force times v1 v2 / c²: magnetism is relativity with the constant coming out right.

The Helmholtz pair: engineering flatness

Two identical coaxial circular coils, equal currents, axial field.

The flatness condition

$$d = R:\ B(0) = \frac{8}{5\sqrt{5}}\frac{\mu_0 N I}{R}$$

$$B(z) = B_0\left[1 - \tfrac{144}{125}\left(\tfrac{z}{R}\right)^4 + \ \ldots\right]$$

Each coil contributes mu0 N I R² / 2(R²+z²)^{3/2} on axis at offset z from its centre. Expanding about the midpoint, odd terms cancel between coils and even ones add; the second derivative vanishes when the half-separation is R/2, i.e. spacing d = R. Then B(0) = 0.716 mu0 N I / R and the leading error is the fourth power of z/R.

WHY / The design rule d = R is a calculus result on the axial formula, not an empirical recipe.

WATCH / The field is flat on axis at the centre only to the stated order; radial uniformity and ends need their own analysis.

$$B''(0) = 0 \ \Leftrightarrow\ d = R$$

Worked design

$$N = 100,\ R = 20\ \text{cm},\ I = 1\ \text{A}:\ B = 0.45\ \text{mT}$$

Flat to 0.1 per cent within |z| < 0.17 R = 3.4 cm: a 7 cm uniform region. To cancel the Earth 45 microtesla inside the pair, drive 0.10 A with the axis along the local field. Coils in opposition can null the field entirely at the cost of the flat region.

WHY / Working one design end-to-end shows the trade between field size, uniformity and current.

WATCH / Uniform to 0.1 per cent is a specification; asking uniform without a tolerance invites wrong answers.

CHECK / Worked design: 100 turns, 20 cm radius, 1 A each gives 0.45 mT flat to 0.1 per cent over about 7 cm.

CHECK / The uniform region spans about 7 cm for 20 cm coils. Each coil contributes half the total at the midpoint by symmetry.

The second method

$$B''(0) = 0 \text{ at } s = R/2$$

inflection of one loop field

Expand each coil field separately: the sum of two identical coils at symmetric positions has no odd terms; the quadratic term carries B double-prime of one loop evaluated at half-separation. A single loop axial field has its inflection at s = R/2, exactly where the partner must sit. Coils in anti-Helmholtz (opposite currents) give a linear gradient instead.

WHY / Anti-Helmholtz is the trap question: opposite currents make a quadrupole used in atom trapping, not a flat field.

WATCH / Collapsing the coils to d = 0 gives one 200-turn loop: B = mu0 N I / R = 0.63 mT - a clean limit check of the 0.716 factor.

Numbers and checks

$$0.716 \to 1\ \ (d \to 0)$$

$\mu_0 NI/R$ is tesla

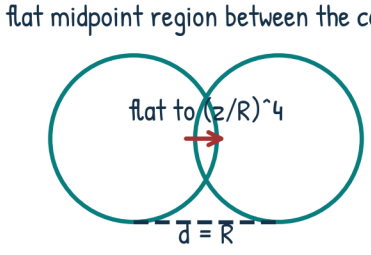
Dimensions of mu0 N I / R are tesla; d to zero returns the single-loop limit; large d splits into two independent loop fields. At the midpoint the field is uniform in the axial direction to fourth order - the entire point of the construction.

WHY / Limit checks verify the constant 8/(5 sqrt 5) = 0.7155 before trusting uniformity claims.

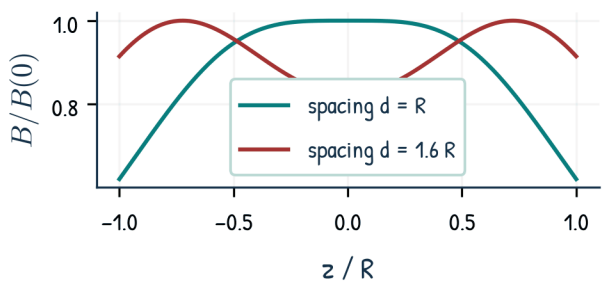
WATCH / Helmholtz uniformity is not spherical uniformity; a large sample sees radial falloff even at the magnetic centre.

CHECK / At the midpoint each coil contributes half the total; the field is uniform in all transverse directions by the cylindrical symmetry.

Helmholtz spacing d = R cancels the quadratic term at the centre



Only d = R flattens the midpoint; wider spacing gives a dip



REMEMBER / Spacing the coils one radius apart cancels the second derivative at the midpoint; uniformity is then fourth order.

The magnetised sphere two ways

Uniform M in a sphere; demagnetising factors; external dipole field.

Rings versus demagnetising factor

$$B_{\text{in}} = \tfrac{2}{3}\mu_0 M$$

$$\mathbf{H} = -N_d \mathbf{M},\ \ N_d = \tfrac{1}{3}$$

Method 1 stacks the bound surface current K = M sin theta as rings and integrates the axial field at the centre. Method 2 uses the general result that inside a uniformly magnetised ellipsoid the demagnetising field is -N_d M with N_x + N_y + N_z = 1; a sphere shares equally, N = 1/3, giving B = mu0(M - M/3). The second method proves the interior is uniform, not just the centre.

WHY / Two methods answer different weaknesses: the ring integral gives one point, the demagnetising argument gives the whole interior.

WATCH / N_d = 1/3 is a sphere result; a long cylinder has N = 0 and a thin plate N = 1.

Worked numbers

$$M = 10^6\ \text{A/m},\ \ R = 1\ \text{cm}$$

$$\mu = 4.19\ \text{A m}^2$$

Interior B = 2/3 mu0 M = 0.84 T. Moment is M times volume: 4 pi R cubed over 3 times M = 4.19 A m². At r = 5R on the axis the dipole field is 6.7e-3 T. A long cylinder of the same M would have 1.26 T inside - larger because its demagnetising field vanishes.

WHY / Comparing shapes at fixed M isolates geometry as the variable; the moment scales with volume in every case.

WATCH / The far-field formula needs r much larger than R; at r = 5R the dipole approximation is already per cent level at the pole.

CHECK / Worked numbers: R = 1 cm, M = 10^6 A/m gives mu = 4.19 A m^2 and B(5R) = 6.7 mT, continuous with the interior at the pole.

Matching the rotating shell

$$K = M\sin\theta$$

$$B_{\text{pole}} = \tfrac{2}{3}\mu_0 M \ \text{continuous}$$

The surface current of a uniformly magnetised sphere equals that of a rotating charged shell with sigma omega R = M. Just outside the pole the dipole field is also 2/3 mu0 M, continuous with the inside as the normal component must be; at the equator outside it is -1/3 mu0 M, a tangential jump mu0 K as required.

WHY / Continuity checks on B_normal and the K jump on tangential H are the field-matching exam skill.

WATCH / Inside the sphere the field is uniform yet the surface current varies with theta; do not expect a solenoid profile.

Sensitivity to shape

$$N_x + N_y + N_z = 1$$

$$N_{\text{sphere}} = \tfrac{1}{3}$$

Ellipsoids are the only shapes with uniform interior field; anything else has non-uniform M response. Laboratory magnetisation measurements therefore quote the sample shape with chi; a powder and a single crystal of the same material can differ.

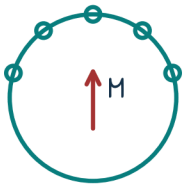
WHY / Demagnetising corrections are the first systematic error in any susceptibility measurement.

WATCH / Reported chi without shape is ambiguous; ask for the demagnetising factor before believing the number.

CHECK / Shape sensitivity: an elongated needle has N_d near zero along its axis and a full mu0 M interior field; a thin disc has the opposite.

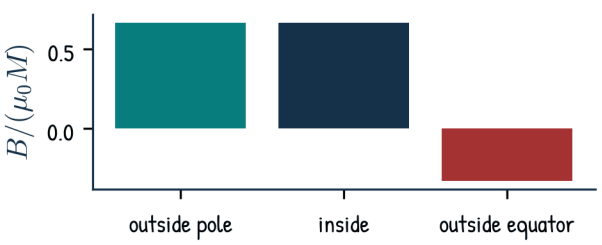
Uniformly magnetised sphere: surface current of a rotating shell

K_b = M sin theta on the surface



inside B = 2/3 mu0 M uniform; dipole outside

Sphere field: uniform inside, dipole outside with a tangential jump



REMEMBER / Interior field two-thirds mu0 M; outside a pure dipole; the demagnetising factor of a sphere is one third.

Magnetic pressure at work

Surface currents confining field; plasma channels; levitated slabs; cosine-theta windings.

Solenoid and pinch

$$P = \frac{B^2}{2\mu_0}$$

$$B_{\text{surface}} = \frac{\mu_0 I}{2\pi a}$$

A 3 T solenoid of 10 cm² bore: 3.6 MPa hoop stress (36 atm) and 3.6 kN pulling the halves together. A 30 kA lightning stroke in a 5 mm channel makes 1.2 T at its surface and pinches itself with 0.57 MPa of inward magnetic pressure, balanced by plasma thermal pressure (the Bennett relation).

WHY / The same formula is outward on a solenoid winding and inward on a plasma column: pressure acts on whatever confines the field.

WATCH / The pinch is unstable (sausage and kink modes); quoting the equilibrium pressure is not a stability proof.

Levitated slab and numbers

$$B = \sqrt{2\mu_0 mg/A}$$

$$1\ \text{T} \rightarrow 4\ \text{atm}$$

A type-I superconductor excludes B entirely and feels the full pressure from outside: supporting 2.0 kg over 4.0 cm² needs 0.35 T at the plate. The Meissner effect makes levitation a stress problem; type-II pinning adds hysteresis and lateral stability.

WHY / Pressure methods give one-line levitation estimates that match image-dipole calculations to a factor of two.

WATCH / A field of 1 T supports a steel slab 5 m thick in principle; practical limits are the superconductor, not the stress.

$$B_{\text{support}} = \sqrt{2\mu_0 mg/A} \approx 0.35\ \text{T}$$

Cosine-theta coil

$$K = K_0 \cos\varphi$$

$$B_{\text{inside}} = \frac{\mu_0 K_0}{2}$$

Let two overlapping current cylinders of the overlap trick approach: their net surface current is K0 cos phi around the cylinder and the interior field stays uniform and transverse. Every accelerator dipole magnet is this construction in superconducting cable, held by steel collars against 27 MPa at 8.3 T.

WHY / The overlap trick taken to its limit designs the LHC dipole: same field theory, engineering scale.

WATCH / Uniform interior requires the full cosine distribution; approximating it with discrete coils leaks multipole errors that steer beams.

Stress checks in numbers

$$P = \frac{B^2}{2\mu_0}$$

$$1\ \text{T} \rightarrow 3.98 \times 10^5\ \text{Pa}$$

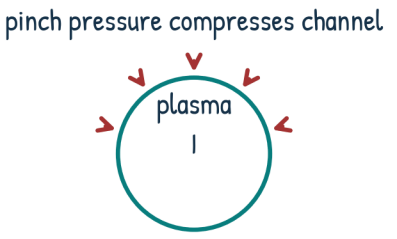
At 3 T the hoop stress is 36 atmospheres; at 8.3 T it is 27 MPa, the LHC collar design point. Remember 1 atm = 1e5 Pa so B squared over 2 mu0 at 1 T is about 4 atmospheres: a two-line sanity check on any magnet mechanical question.

WHY / A pressure-of-field table converts stress answers instantly and catches unit slips in Pa versus atm.

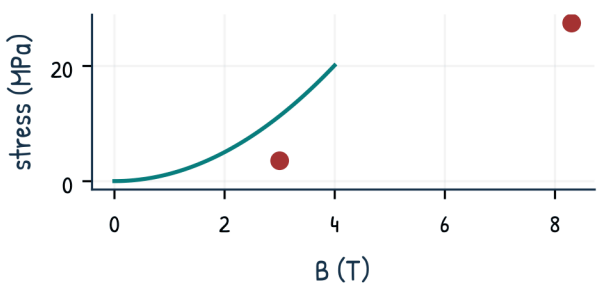
WATCH / Stress acts on the current distribution, not on the vacuum; quoting B squared over 2 mu0 as a pressure in empty space needs the confining current named.

CHECK / Hoop stress in a 3 T pulse coil is 3.6 MPa in copper; pulsed machines survive milliseconds of it and then yield plastically.

Current channel pinched by its own magnetic pressure



Hoop stress from B squared: 3.6 MPa at 3 T, 27 MPa at 8.3 T



REMEMBER / B²/2 mu0 is engineering: hoop stress, pinch, levitation and accelerator dipoles all obey it.

The cycloid family and Fermi acceleration

Crossed uniform fields; then slowly moving magnetic mirrors.

Classify by drifting frame

$$v_d = E/B, \quad r = \frac{m|\mathbf{v}_0 - \mathbf{v}_d|}{qB}$$

$$\text{wheel radius } r_c = \frac{mE}{qB^2}$$

In the drifting frame the particle circles at speed |v0 - vd|; the lab path is that circle carried along at vd. v0 = vd is a straight selector line; |v0 - vd| = vd is a cusped cycloid; larger gives loops (prolate), smaller curtate arches. Protons in E = 2e3 V/m, B = 0.020 T: vd = 1e5 m/s, r_c = 5.2 cm.

WHY / The drifting-frame construction classifies every initial condition in one picture; solving lab equations of motion reproduces it.

WATCH / Cusped, curtate and prolate refer to the gyration radius relative to the wheel, not to different forces.

$$\mathbf{v} = \mathbf{v}_d + \mathbf{v}_{\text{circle}}$$

Energy across one loop

$$v_{\text{top}} = 3v_d \text{ in a loop}$$

$$\Delta K = qE \cdot 2r$$

Energy conservation over one arch gives the top speed from qE times height 2r; in a prolate loop the bottom speed exceeds 3vd and the top falls to vd, checked by half m v² differences. The drift is the same for an electron; only the circle size changes by the mass ratio.

WHY / Energy accounting verifies trajectory classification without integrating equations of motion.

WATCH / The proton and electron arch heights differ by 1836 even in identical fields; never transfer one orbit dimension to the other.

CHECK / Over one loop the E field does net work qE times the drift distance; the magnetic force does none, so the arch height is set by E alone.

Fermi acceleration

$$\frac{\Delta K}{K} \approx \frac{4V}{v}$$

$$N \approx 0.21\frac{v}{V} \text{ to double}$$

A particle reflecting from an approaching mirror gains 2V per head-on bounce in speed, hence about 4V/v in energy. Doubling energy takes about 0.21 v/V reflections: for cosmic rays on 10 km/s clouds that is 6000 bounces over 6000 years per light-year, and receding mirrors cancel half the gain at second order - shock acceleration is the modern resolution.

WHY / First-order (head-on shocks) and second-order (random clouds) Fermi acceleration differ in efficiency, not in principle.

WATCH / The adiabatic invariant p_parallel times path length gives the same gain law for slow compression; do not mix adiabatic and impulsive regimes in one estimate.

Cycloid numbers worked

$$E = 2.0 \times 10^3 \text{ V/m}, \quad B = 0.020 \text{ T}$$

$$v_d = 1.0 \times 10^5 \text{ m/s}$$

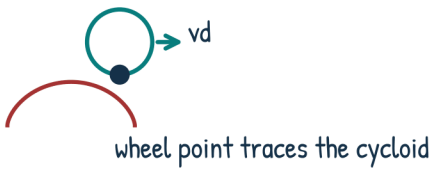
Protons: wheel radius r_c = m v_d / eB = 5.2 cm and gyro period 3.3 microseconds. Launch along +x at vd gives a straight line; at 2 vd a cusped cycloid; from rest the classic arch of height 10.4 cm; at minus vd prolate loops with instantaneous reverse motion.

WHY / Four initial conditions, one construction: classify with |v0 - vd| before integrating anything.

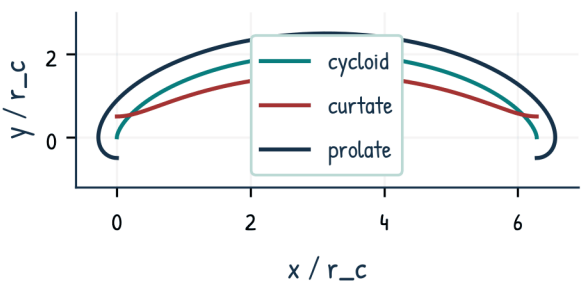
WATCH / In a prolate loop the instantaneous velocity reverses at the bottom of each loop; do not assume monotone forward motion.

CHECK / Worked cycloid: E = 10^4 V/m, B = 0.1 T gives r_c = 5.7 micrometres, v_max = 2e5 m/s and cusps every 0.36 ns for an electron.

Rolling wheel picture: the gyration centre drifts at E cross B over B squared



Wheel-point distance sets cycloid family: curtate, cusped or looping



REMEMBER / Every crossed-field orbit is a rolling wheel; every reflection off an approaching mirror gains energy.

Three clocks and a synchrotron

Trapped particles in dipole fields; ultrarelativistic electrons in ramps.

Gyration, bounce, drift

$$T_c = \frac{2\pi m}{eB}$$

$$T_d \approx \frac{2\pi e B r^2}{3K}$$

A 1 MeV proton at L = 2 in the Earth field (3.9 microtesla): gyration 17 ms with a 37 km radius, bounce between mirrors about 2 s, gradient drift around the Earth about 1300 s westward. The three timescales separate by two orders each, making each motion an adiabatic invariant of the next.

WHY / Hierarchy of clocks is what makes guiding-centre theory work; check the separation before applying invariants.

WATCH / Curvature drift adds to gradient drift for particles with parallel motion; quote which drift the period includes.

$$T_g : T_b : T_d \sim 10^{-2} : 10^{-1} : 1 \text{ s}$$

Drift period scaling

$$T_d \propto \frac{1}{K}$$

$$\text{ring current of MA}$$

Higher-energy particles drift faster and circle sooner: a 10 MeV proton takes 2 minutes, a 10 keV one 1.5 days. The trapped population carries a westward ring current of a few mega-amperes, the geomagnetic storm signature.

WHY / Energy-dependent drift times explain why injections of energetic particles disturb compasses within minutes.

WATCH / Dipole field model and local time effects give order-of-magnitude periods; exact drift shells need full field models.

CHECK / The drift clock at L = 2: protons drift east, electrons west, both circling the Earth in minutes to hours by charge sign.

The synchrotron

$$f = \frac{eB}{2\pi\gamma m}$$

$$\Delta E_{\text{turn}} = 88.5 \text{ keV} \frac{E^4[\text{GeV}]}{r[\text{m}]}$$

A 1 GeV electron in 1 T has orbit radius 3.3 m and revolution 14.3 MHz; the cyclotron formula would say 28 GHz - wrong by gamma = 1958. The synchrotron holds r fixed by ramping B with momentum and tunes the cavity towards c/2 pi r. Electrons radiate 27 keV per turn at that energy and must be replenished.

WHY / Synchrotron radiation sets the energy ceiling of electron machines and is itself a light source.

WATCH / Radiation loss scales as the fourth power of energy: small energy increases, large losses. Classical cyclotron answers for electrons are always wrong.

$$f = qB/2\pi\gamma m$$

Detector-scale numbers

$$p[\text{GeV}/c] = 0.3\, B[\text{T}]\, r[\text{m}]$$

$$E_e \text{ radiates, } E_p \text{ less}$$

The 300 Br rule in GeV units is the workhorse of collider design. Proton machines radiate far less at the same energy because the loss scales as 1/m⁴ at fixed energy; this is why the LHC collides protons while light sources exploit electron radiation.

WHY / Knowing which particle radiates decides accelerator architecture and detector backgrounds.

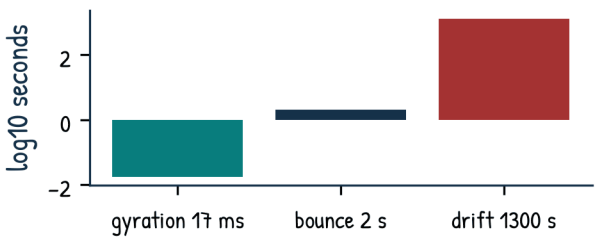
WATCH / At fixed radius and B, momentum is fixed but energy depends on mass; always state which quantity the machine holds fixed.

CHECK / Detector scales: 1 GeV/c in 1 T is a 3.3 m radius and a 14.3 MHz revolution; at that energy the electron is ultrarelativistic and radiates.

One trapped particle keeps three clocks: gyration, bounce, drift



Three clocks separated by two orders each make adiabatic invariants



REMEMBER / Radiation-belt particles keep three well-separated clocks; a synchrotron abandons the fixed cyclotron frequency.

Two carriers, levitation and why iron beats water

Mixed carriers in semiconductors; image dipoles over perfect diamagnets; susceptibility estimates.

Two-carrier Hall

$$R_H = \frac{p\mu_p^2 - n\mu_n^2}{e(p\mu_p + n\mu_n)^2}$$

With electrons and holes both mobile, the transverse current balance weights each by the square of its mobility: R_H vanishes when n mu_n² = p mu_p² and flips sign beyond. Germanium at room temperature is p-type by count yet Hall-positive; heated to n = p it becomes Hall-negative, R_H = -7.2e-3 m³/C.

WHY / Mobility weighting, not carrier counting, decides the Hall sign: fast electrons dominate slow holes of equal number.

WATCH / The vanishing line n/p = (mu_p/mu_n)² is where R_H passes through zero; near it the simple one-carrier formula is doubly wrong.

$$R_H = \frac{n_h\mu_h^2 - n_e\mu_e^2}{e(n_h\mu_h + n_e\mu_e)^2}$$

Levitation by images

$$h = \left(\frac{3\mu_0\mu^2}{32\pi mg}\right)^{1/4}$$

$$F = \frac{3\mu_0\mu^2}{2\pi s^4}$$

A perfect diamagnet reflects a dipole into an antiparallel image at the mirror depth; coaxial antiparallel dipoles at separation 2h repel as 3 mu0 mu² / 32 pi h⁴. A 1 cm³ NdFeB cube of moment 1 A m² floats 2.7 cm above a superconductor; doubling its weight only lowers h to 2.3 cm.

WHY / The quarter-power makes levitation forgiving: masses and moments change heights slowly.

WATCH / Type-II pinning adds lateral stability that the ideal image calculation cannot give; ideal Meissner levitation is laterally unstable.

$$F = \frac{3\mu_0\mu^2}{2\pi(2h)^4}$$

Iron versus water numbers

$$\chi_{\text{water}} \approx -9 \times 10^{-6}$$

$$k_B T_c \approx zJ/3$$

Dipole coupling between atomic moments is 0.04 K worth of energy - nothing at room temperature. Exchange J about 0.05 eV with z = 8 neighbours gives Tc near 1500 K, the right order for iron 1043 K. Water diamagnetism from the Larmor formula with Z = 10 and <r²> = (0.07 nm)² gives -9.5e-6 against the measured -9.0e-6.

WHY / Three order-of-magnitude estimates in one problem: rule out the wrong mechanism, estimate the right one, verify the weak universal effect.

WATCH / Classical Larmor diamagnetism gets the right scale only with a sensible <r²>; the agreement for water is partly fortunate in that choice.

Perfect versus Larmor diamagnet

$$\chi = -1 \ \text{(Meissner)}$$

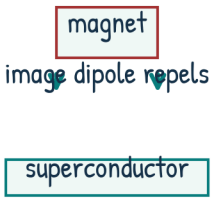
$$\chi_{\text{Larmor}} \sim -10^{-5}$$

A superconductor screens B completely with surface currents and floats magnets centimetres high; ordinary diamagnets reduce the field by parts per million and levitate nothing visible. Both are negative susceptibilities; only one reaches the Meissner limit. Levitation estimates use image dipoles or B squared over 2 mu0 stress.

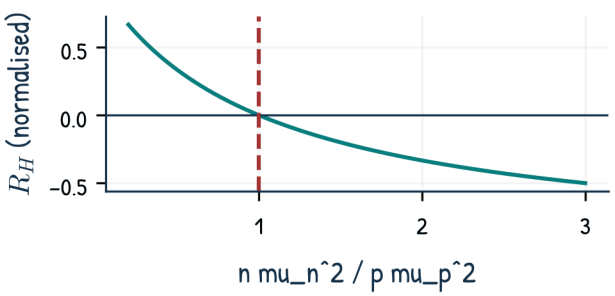
WHY / Separating the two mechanisms prevents the classic error of using chi = -1e-5 in a levitation estimate.

WATCH / Type-II superconductors trap flux and pin vortices: ideal Meissner image formulas describe type-I or reversible type-II only.

Levitation: a perfect diamagnet reflects the dipole into an opposite image



Two-carrier Hall sign flips when n mu_n squared overtakes p mu_p squared



REMEMBER / Mixed carriers can invert the Hall sign; levitation heights follow from image forces; iron orders by exchange.

Estimates, Gauss method and model limits

Order-unity geometry factors; dipole model of the Earth; classical magnetostatics assumptions.

Gauss absolute measurement

$$\mu B_H = \frac{4\pi^2 I_m}{T^2}, \qquad \frac{\mu}{B_H} = \frac{\tan\theta\, d^3}{2\times 10^{-7}}$$

Oscillating a magnet of moment of inertia I_m gives the product mu B_H; deflecting a compass from distance d gives the ratio mu / B_H. I_m = 6e-6 kg m², T = 4.0 s, d = 20 cm, theta = 30 degrees give mu = 0.58 A m² and B_H = 25 microtesla: a field in mechanical units at last.

WHY / Two measurements, two unknowns: the structure of the method is the physics, not the arithmetic.

WATCH / The deflection formula needs the magnet far enough for its dipole field and the compass in its meridian; both are standard idealisations.

$$\mu = 0.58\ \mathrm{A\,m^2},\quad B_H = 25\ \mu\mathrm{T}$$

CHECK / Gauss method from T = 4.0 s, I_m = 6e-6 kg m^2 and a 30-degree compass deflection at 20 cm reproduces the dipole moment.

Estimates that teach

$$I_{\text{core}} \sim 2\times 10^9\ \mathrm{A}$$

$$B_{\text{heart}} \sim 100\ \mathrm{pT}$$

A core-sized current loop needs two billion amperes, hence a dynamo. A compass within 10 cm of a 0.25 A wire errs a degree; a car battery lead must sit a metre away. The heart current dipole of 1e-5 A m produces 100 pT at 10 cm, measurable only with SQUIDs near the flux-quantum noise floor.

WHY / Each estimate answers a question a curious examiner can ask; the physics is the scaling, the number is the check.

WATCH / Bioelectric current dipoles are ampere-metres with 1/r² fields; orbital magnetic dipoles are ampere-metre-squared with 1/r³ fields. Mixing the two powers is the classic slip.

Where the model breaks

$$\gamma = qB \text{ failures: } f = \frac{eB}{2\pi\gamma m}$$

$$\chi = -1\ \text{(superconductor)}$$

Relativity moves the cyclotron frequency by 1/gamma; circling charges radiate; Landau quantisation and g = 2 are quantum; iron saturates at 2.1 T and its mu_r is a loop slope; changing flux induces electric fields - the next module; plasmas add collective effects and instabilities.

WHY / Listing failure modes is the mature end of a physics chapter: knowing when not to trust the formulas.

WATCH / Superconductivity and exchange are quantum from the ground up; classical analogies there are metaphors, not derivations.

CHECK / The model breaks for: relativistic frequencies, radiating orbits, quantum spin g = 2, saturated iron, and changing fields needing induction.

Consistency identities

$$\tan\theta_{\text{dip}} = 2\tan\lambda$$

$$B = B_{\text{eq}}\sqrt{1+3\sin^2\lambda}$$

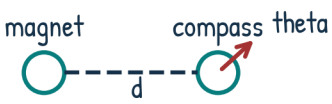
A station with dip 60 degrees sits at magnetic latitude 41 degrees in the dipole model; its total field is 1.5 B_eq. Checking both identities on any station table separates model agreement from local anomaly data. The dipole moment from B_eq = 30 microtesla gives 7.8e22 A m².

WHY / The dip-latitude identity is a free consistency check in every Earth-field question.

WATCH / Dipole geometry assumes a centred, aligned dipole; the 11 degree tilt and secular drift spoil precision but not the estimates.

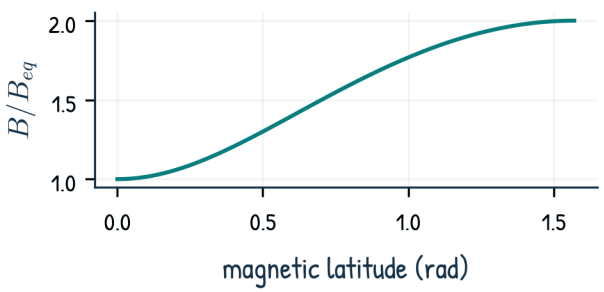
CHECK / Dip and latitude identities: dip 60 degrees sits at magnetic latitude 41 degrees; total B = 1.5 B_eq there — verify both on any station table.

Absolute measurement: magnet oscillation plus compass deflection at distance d



oscillation: product; deflection: ratio

Dipole field strength grows from equator to pole by up to a factor two



REMEMBER / Estimates are arguments: each one shows what the model can and cannot do.

The Faraday disc and the flux rule limits

Conducting disc spinning in axial B with brushes; sliding contacts; axisymmetric magnets.

The disc from the Lorentz force

$$\mathcal{E} = \tfrac{1}{2} B \omega a^2$$

An element at radius r moves at omega r and feels omega r B along the radius; integrate from axle to rim: half B omega a squared. A 10 cm disc at 300 rad/s in 0.5 T gives 0.75 V and, into a milliohm, 750 A with braking torque one half I B a squared. The homopolar generator: enormous current at a fraction of a volt.

WHY / No circuit was needed: the physics is the force on the carriers in the moving copper.

WATCH / The torque method gives I B a squared / 2 = E I / omega identically; energy conservation is built into the geometry.

$$\tau = \tfrac{1}{2} I B a^2 = \frac{\mathcal{E} I}{\omega}$$

The flux rule done right

$$\Phi = B \times \text{swept area}$$

Take as the circuit the radial line of copper that rotates with the disc, closed through the brush and axle: its area grows at one half a squared omega and the flux rule reproduces the EMF. Choosing the FIXED radial line through the brush gives a constant flux and the wrong answer, zero: the paradox is a mis-chosen circuit.

WHY / Choose a circuit whose boundary is the moving material and the flux rule agrees with the charges exactly.

WATCH / A wheel rolling on rails with a rim contact, or a contact jumping wires, breaks the premise; follow the charges there.

CHECK / The disc is the classic flux-rule embarrassment and the Lorentz-force easy victory.

Spin the magnet instead

$$\mathcal{E} = 0 \ \text{(magnet only)}$$

An axisymmetric magnet has the same field whether or not it spins; with the disc still no charges move through it and the EMF is zero. If both spin together the disc charges move through the same field and the EMF is unchanged. Only the CONDUCTOR motion counts, which is what follow the charges means.

WHY / The unipolar inductor with brushes on a spinning magnetised cylinder does generate: the magnet material carries its own charges.

WATCH / Exam questions switch which part rotates; answer from the charges, not from the picture of a rotating field.

CHECK / Faraday 1831 disc: 0.05 T, 15 cm, a few turns per second: tens of millivolts, enough for his galvanometer.

The sliding contact family

$$d\mathcal{E} = (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

A loop whose closing contact jumps conductors changes its flux discontinuously with no EMF at all; no charge moved through a field. Every member of the paradox family is resolved by computing the force on the material that actually moves and integrating along it.

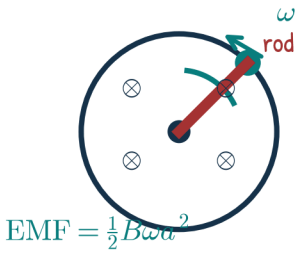
WHY / One rule covers the family: the Lorentz force is the law; the flux rule is a theorem with hypotheses.

WATCH / Where in doubt, write the v cross B integral before the flux; it never lies.

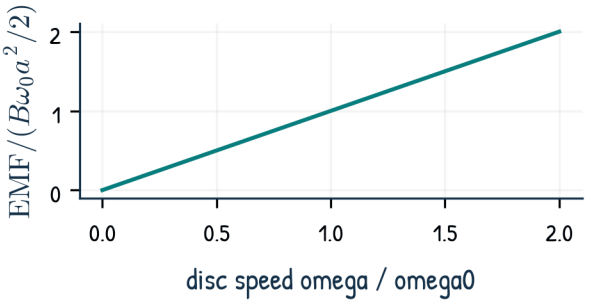
CHECK / A radial line rotating with the disc gives the swept-area rate one half a squared omega: both methods agree.

CHECK / A contact jumping conductors changes flux discontinuously with no EMF: no charge moved through a field.

Faraday disc: every radius is a rod, all in parallel



The disc EMF is linear in spin rate: half B omega a squared



REMEMBER / Follow the charges that actually move; the flux rule is a shortcut for loops that carry their material.

The betatron from canonical momentum

Axisymmetric changing flux; electrons starting from rest; classical mechanics with a vector potential.

Canonical angular momentum

$$mvr - \frac{e\Phi}{2\pi} = \text{const} = 0$$

The induced field is azimuthal; its torque about the axis changes the orbital angular momentum at the rate e over 2 pi times the flux rate. For electrons at rest before the flux existed the constant is zero: at every instant p = e Phi / 2 pi r. This is Noether theorem for rotational symmetry.

WHY / A conservation law derived once replaces the rate comparison; the condition holds at all times, not just on average.

WATCH / A non-axisymmetric field breaks the conservation; betatrons are built round it for exactly this reason.

$$p = \frac{e\Phi}{2\pi r} = eB_{\text{orb}}r \;\Rightarrow\; B_{\text{orb}} = \tfrac{1}{2}\langle B\rangle$$

CHECK / The conserved quantity is m v r plus e A_phi r: Noether theorem for rotational symmetry. A non-axisymmetric field would break it.

The orbit condition recovered

$$B_{\text{orb}}r = \frac{\Phi}{2\pi r}$$

Setting p = e B_orb r against p = e Phi / 2 pi r gives B_orb = Phi / 2 pi r squared = one half the average field: the 2:1 condition at every instant with no differentiation. The conserved quantity is m v r plus e A_phi r with A_phi the azimuthal vector potential.

WHY / The vector potential is doing real work here; the canonical momentum is the deeper reason the machine functions.

WATCH / The rate argument and the conservation argument must agree; if they do not, a sign in the torque is wrong.

CHECK / Dimensions of e Phi / 2 pi r are kg m/s: coulomb weber per metre is momentum.

CHECK / Differentiate p = e B r directly: the force is purely radial only while the orbit field is frozen.

Radial stability

$$B \propto r^{-n}, \qquad 0 < n < 1$$

Displace the electron outward at fixed p: the centripetal demand falls as 1/r while the available force evB falls as r^-n; restoring needs n below 1. Vertical stability needs field lines bowing outward: n above 0. The shaping that gives 2:1 across the orbit can satisfy both.

WHY / Weak focusing lives in one exponent window; the pole faces are shaped to hold it.

WATCH / A uniform field gives n = 0: vertical stability fails even though the 2:1 ratio holds trivially.

$$p = eB_{\text{orb}}R = 36\ \text{MeV}/c \text{ for } R = 0.3\ \text{m},\ 0.4\ \text{T}$$

CHECK / A uniform field gives n = 0: vertical stability fails even though the 2:1 ratio holds trivially. Pole shaping holds the window.

Numbers and checks

$$\Delta\Phi = \pi R^2\langle B\rangle = 0.226\ \text{Wb}$$

R = 0.3 m with B_orb reaching 0.4 T: p c = 36 MeV and the total flux change 0.226 Wb. The betatron works for electrons precisely because the argument used p, valid at any speed; the fixed-frequency cyclotron fails as gamma grows.

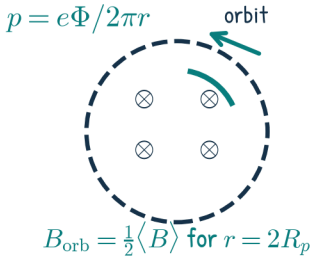
WHY / Relativistic safety is a property of the formulation: p = eBR never mentioned mass.

WATCH / Check the flux integral against 2 pi R p/e; the two expressions for Delta Phi must close the audit.

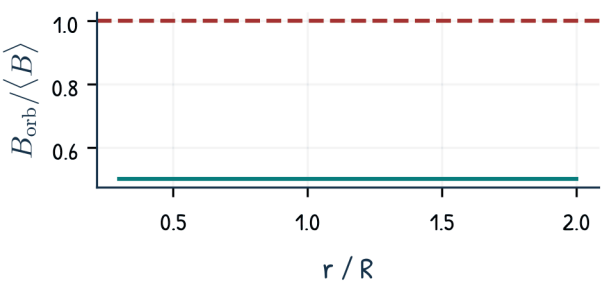
CHECK / The 2:1 machine at R = 0.5 m reached 75 MeV with 157 eV per turn over 4.8e5 turns.

CHECK / A 36 MeV orbit at 0.5 T has radius 0.24 m and circumference 1.5 m: table-top sized for its energy.

Canonical momentum in a symmetric betatron keeps the orbit fixed



Conserved canonical momentum forces B_orb = half the average field at every radius



REMEMBER / The 2:1 condition follows from a conservation law, not rates: the canonical angular momentum is constant.

The magnet in the tube and braking oscillators

Point dipole falling in a thin conducting tube; copper bob on a string between magnet poles.

The tube derivation

$$F = \frac{45}{1024}\frac{\mu_0^2 \mu^2 \sigma t}{a^4} v$$

Each ring of the tube sees dPhi/dt = v dPhi/dz from the falling dipole; integrating the ring dissipation with the dipole flux integral gives the drag coefficient K. For 1 A m^2 in a 1 cm, 1 mm-wall copper tube: K = 0.42 N s/m, terminal speed mg/K = 0.18 m/s, reached in 18 ms.

WHY / The scalings are exact and useful: drag one over a^4, one over sigma t, and mass over moment squared.

WATCH / Point-dipole and thin-wall assumptions agree with teaching-lab measurements to tens of per cent.

$$v_t = \frac{mg}{K} = 0.18 \text{ m/s for } 7.5 \text{ g}$$

CHECK / Measure terminal speed at two fields: v falls as one over B squared if the drag model is right. Integrating the finite magnet length improves agreement to a few per cent.

Energy audit and limits

$$mgv_t = Kv_t^2$$

At terminal speed the gravitational power equals total i squared R in the rings: the audit that rederives the same K. A plastic tube gives free fall; halving the radius slows the magnet sixteen times; a thicker wall breaks the thin-wall formula but not the physics.

WHY / Two methods, one answer: the energy audit is the cross-check every derivation deserves.

WATCH / A real 1 cm magnet near a 1 cm wall is not a point dipole; integrating the finite length improves agreement to a few per cent.

CHECK / Terminal speed of tenths of a metre per second is Lenz law you can hold in a demonstration tube.

CHECK / The energy audit is mechanical: m g h equals the heat in the tube wall, with no free parameters.

Braking as damping

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0, \qquad 2\gamma = \frac{b}{m}$$

A copper bob on a string in a magnet gap is a damped pendulum with b = k sigma B squared V. Critical damping at b = 2 m omega0 happens near 0.08 T; above a tenth of a tesla the bob creeps back without overshooting, and at 0.5 T it stops within one swing.

WHY / The demonstration sits far past critical damping; slotted bobs drop k by twenty and swing for minutes.

WATCH / Doubling the field quadruples the damping: B squared scaling decides the regime.

$$B_c = \sqrt{\frac{2m\omega_0}{k\sigma V}} \approx 0.08 \text{ T}$$

CHECK / Critical damping near 0.08 T: above a tenth of a tesla the bob creeps back without overshooting. Doubling the field quadruples the damping.

The two regimes

$$Q = \frac{\omega_0}{2\gamma}, \qquad b \propto B^2$$

Below critical the amplitude decays exponentially with visible swings and a Q of order unity to ten; above, the return is aperiodic. At 0.02 T the bob keeps about 8 visible swings; at 0.5 T it is forty times critical.

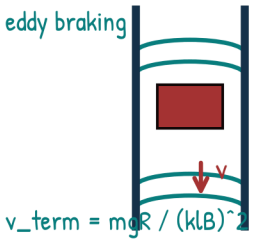
WHY / One parameter b/B squared spans the whole behaviour; classify before solving.

WATCH / The slotted bob has k near 0.01 of the solid value: the lamination principle as a control experiment.

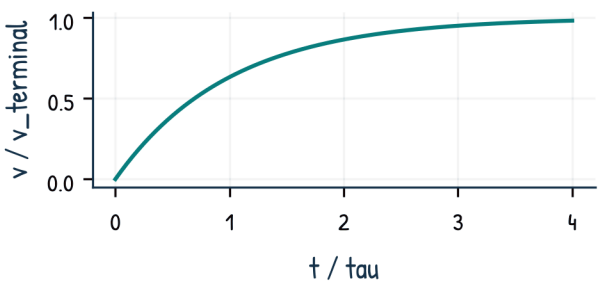
CHECK / The pendulum period at B = 0 is 2 pi root(l/g) = 1.4 s: the undamped limit checks the constants.

CHECK / At 0.02 T the bob keeps about 8 visible swings; at 0.5 T it is forty times critical. The slotted bob drops k twentyfold and swings for minutes.

A magnet falls through a conducting tube at a crawl: terminal velocity law



The falling magnet approaches a terminal speed of tenths of a metre per second



REMEMBER / Eddy drag laws can be derived in closed form; the falling magnet reaches a tenth of a metre per second.

Flux conservation in superconducting loops

Zero-resistance loops; switching transients; isolated rings with persistent current.

The switched pair

$$I = \frac{L_1 I_0}{L_1 + L_2}, \qquad \Delta U = \frac{L_2}{L_1 + L_2} U_0$$

Switching a current-carrying L1 across an uncharged L2 conserves total linkage and loses a fraction of the energy in the arc, however small the resistances. 10 mH at 3 A into 20 mH: common current 1 A, energy 45 to 15 mJ, and 30 mJ leaves in the spark and wire resistance during equalisation.

WHY / The amount of loss is fixed by conservation; only the mechanism depends on R.

WATCH / The inductive twin of joining two charged capacitors; the same audit structure serves all three.

CHECK / L2 to zero loses nothing; L2 to infinity loses everything: both limits close the formula.

CHECK / The lost energy fraction is L2 over L1 plus L2: the two-capacitor paradox with C traded for inverse L.

The squashed ring

$$I_2 = \frac{L_1}{L_2} I_0, \qquad U = \frac{\Lambda^2}{2L}$$

A superconducting ring carrying 100 A at 1 microhenry squashed to 0.5 microhenry doubles its current to 200 A and its energy rises 5 to 10 mJ: the extra energy is the mechanical work done against the magnetic pressure resisting the squash. At fixed flux U = Lambda squared / 2L and the force is one half I squared dL/dx.

WHY / The ring pushes toward larger L; releasing it springs back and returns the work.

WATCH / The force formula is the same as at fixed current, but the energy ledger differs; quote the constraint before computing.

$$F = \tfrac{1}{2} I^2 \tfrac{dL}{dx} \ \text{(fixed flux, same force)}$$

The floating magnet

$$\Lambda = \Phi_{\text{ext}} + LI = \text{const}$$

A magnet dropped at a superconducting ring never gets through: the induced current grows to cancel the extra flux and the repulsion grows without limit. The image is perfect diamagnetism in circuit language: the loop enforces zero flux change by whatever current it takes.

WHY / Flux conservation is the circuit form of chi = -1.

WATCH / Real rings have resistance and the magnet passes eventually; the levitation time is the L/R time.

CHECK / Flux quantum counting: 1e-4 Wb in the worked ring is 5e10 quanta: classical to any precision.

CHECK / The ring pushes toward larger L; releasing it springs back and returns the work. The force formula matches the fixed-current one, but the ledger differs.

When conservation fails

$$t \ll L/R \ \text{conserves}; \qquad t \gtrsim L/R \ \text{decays}$$

With resistance the flux is conserved only for times short compared with L/R; afterwards Faraday with R applies and the current decays. The initial-condition rule that an inductor current cannot jump is the t = 0 limit of exactly this statement.

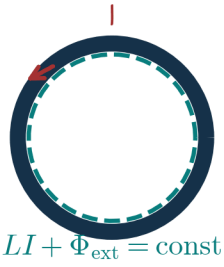
WHY / One conservation law covers both the superconducting ideal and the short-time transient.

WATCH / Do not apply flux conservation to a circuit that has had time to decay; check the time scale first.

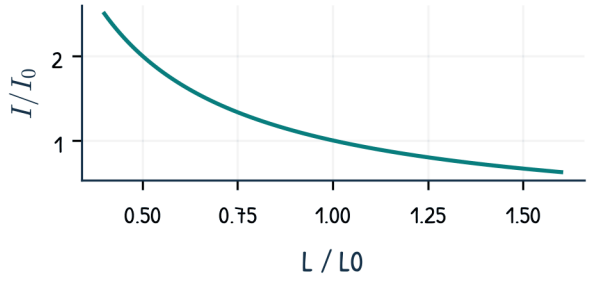
CHECK / A 1 milliohm, 10 microhenry loop has tau = 10 ms: conservation is a short-time tool even in copper.

CHECK / Real rings have resistance and the magnet passes eventually; the levitation time is the L / R time. Flux conservation is the circuit form of chi = -1.

Superconducting ring: flux is frozen; current appears to hold it



Flux conservation: squash the ring and the current rises as 1/L while energy rises



REMEMBER / Flux linkage is the momentum of a superconducting loop; energy is not conserved and the switch pays the difference.

The tether and the coil launcher

Satellite tether in the geomagnetic field; pulse coil launching an aluminium ring.

The electrodynamic tether

$$\mathcal{E} = BLv = 4.6 \text{ kV}$$

A 20 km tether at 7.7 km/s through 30 microtesla gives 4.6 kV; with plasma contactors at 0.5 A the force is I L B = 0.3 N opposing the orbit, extracting 2.3 kW electrical from the orbital energy. The orbit decays about 1 m/s, 90 km per day: a propellantless deorbit device.

WHY / A 1 A current in a 20 km tether near the equator feels about 0.6 N in the 30 microtesla field: thrust without propellant.

WATCH / Over the poles the field is vertical and the EMF vanishes; the geometry is equatorial.

$$F = ILB = 0.30 \text{ N}, \qquad P = \mathcal{E}I = 2.3 \text{ kW}$$

CHECK / The 1996 Tethered Satellite measured 3.5 kV and up to 0.5 A on 19.7 km before the tether broke.

The coil launcher

$$I_2 = -\frac{M}{L_2} I_1 = -4.9 \text{ kA}$$

A 50 A pulse in a 200-turn coil couples to a resting aluminium ring with M = 20 microhenry; during a short pulse the ring conserves its flux and drives 4.9 kA opposing. The force I1 I2 dM/dz about 100 N upward launches the 20 g ring at 5 m/s to 1.2 m.

WHY / The launcher is 1.5 per cent efficient: the rest heats the ring through its own resistance.

WATCH / The pulse must be short compared with the ring time constant or the current decays mid-pulse and the force halves.

$$F = I_1 I_2 \frac{dM}{dz} \approx 98 \text{ N}, \qquad v \approx 5 \text{ m/s}$$

CHECK / The pulse must be short compared with the ring time constant or the current decays mid-pulse and the force halves. A slit ring does not jump.

Energy split and controls

$$\tfrac{1}{2} m v^2 = 0.24 \text{ J vs } 16 \text{ J heat}$$

Kinetic energy is 0.24 J against 16 J of ring heating: the copper resistance swallows the pulse energy. A slit ring carries no current and does not jump: the standard control proving induction is the mechanism. The AC version with an iron core gives steady repulsion from the lagging ring current.

WHY / The efficiency estimate is the honest part of the design: most energy is heat, as in every induction launch.

WATCH / The ring current lags by nearly 90 degrees in the AC case; the average force is repulsive but oscillates at 2f.

CHECK / Dimensions of I1 I2 dM/dz are amperes squared henry per metre: newtons.

CHECK / Kinetic 0.24 J against 16 J of ring heat: about 1.5 per cent efficient.

Scaling the launch

$$v \propto \frac{M I_1}{L_2 m} \Delta t$$

Faster pulses need lower ring inductance or higher coupling; the AC demonstration trades peak force for simplicity. Real launchers stack stages: each coil fires as the ring passes, and the ring heats progressively.

WHY / Impulse divided by mass is the launch speed; the coil geometry sets both M and its gradient.

WATCH / A 1 ms pulse against a 0.3 ms ring time constant is on the edge: solve the coupled equation rather than assume conservation.

CHECK / The ring velocity estimate 2 to 3 m/s with a longer pulse matches the demonstration.

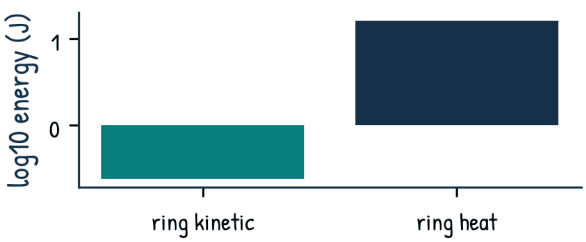
CHECK / Impulse divided by mass is the launch speed. Real launchers stack stages fired as the ring passes; the ring heats progressively.

Coil launcher: mutual inductance gradient drives the ring skyward



fast pulse launches the ring: 1.5 % efficient

The coil launcher is 1.5 per cent efficient: most pulse energy heats the ring



REMEMBER / Motional EMF in orbit is kilovolts; the coil launcher is mutual inductance with an energy ledger.

Switched inductors and the eddy drag law

Two inductors joined by a switch; a magnet passing a coil of inductance L and resistance R.

Two inductors and a switch

$$I = \frac{L_1 I_0}{L_1 + L_2} = 1.0\ \text{A}$$

10 mH carrying 3 A switched across 20 mH: common current 1 A and energy 45 to 15 mJ; the 30 mJ lost is the capacitors-note loss in inductive dress. The amount is fixed by flux conservation; the mechanism is the arc and whatever resistance exists.

WHY / State both limits: L2 to zero loses nothing, L2 to infinity loses everything.

WATCH / Deliberate resistance in the loop spreads the same loss over time and makes the switching gentle.

CHECK / The lost fraction is L2 / (L1 + L2) of the initial energy: check both limits before trusting it.

CHECK / Deliberate resistance in the loop spreads the same loss over time and makes the switching gentle. The lost fraction is L2 over L1 + L2.

The slow passage

$$F = -\frac{\Phi'^2}{R} v \ \text{(low speed)}$$

A magnet passing a coil drives EMF = -v Phi prime; when the passage is slow compared with L/R the current follows the resistance and the drag is proportional to speed: the eddy-brake law of the standard sheets. The force equals the EMF power divided by v, an energy identity.

WHY / The linear regime covers brakes, tubes and hobs because their loop time constants are microseconds.

WATCH / The sign: the force always opposes the relative motion; Phi prime squared guarantees it.

$$Fv = \mathcal{E} i = \frac{(v\Phi')^2}{R}$$

CHECK / Low-speed drag is v times (dPhi/dx) squared over R: the brake law in one line.

The fast passage

$$Li = -\Delta\Phi \ \Rightarrow \ F \text{ independent of } v$$

When the passage is fast compared with L/R the coil conserves flux; the current is set by the flux change alone and the integrated impulse is zero: an elastic bounce, no drag and no heat. In between, the drag peaks when omega tau = 1 and falls again.

WHY / The superconducting limit is no drag at all; real brakes work only below the peak speed.

WATCH / A thick copper block with millisecond tau is a bad brake at high speed; thin rings keep tau short.

CHECK / Peak-drag speed for a 1 cm ring of 1 mm copper is about 30 m/s: far above the falling magnet 0.2 m/s.

CHECK / The peak sits at v = w R over L; designers size the loop so the service speed lies left of it.

The intermediate law

$$F \propto \frac{v}{1+\omega^2\tau^2}, \qquad \omega \sim v/w$$

For a sinusoidal passage the dissipative part of the force rises linearly at low speed, peaks at omega tau = 1 and falls as 1/v beyond: the general picture behind every induction brake. The full solution is the driven RL transient with the magnet as source.

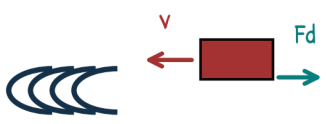
WHY / One curve covers brake design: know where on it your device sits.

WATCH / The falling magnet and the pendulum sit safely in the linear regime; high-speed maglev does not.

CHECK / At the peak the drag equals the low-speed law extrapolated to v = wR/L: continuous interpolation.

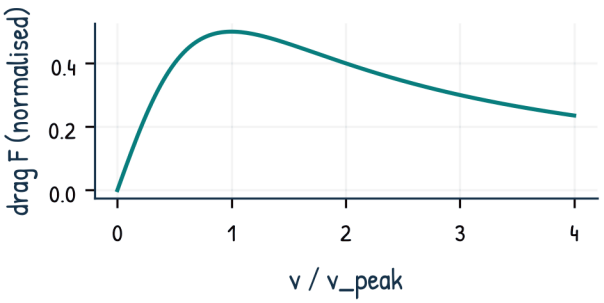
CHECK / One curve covers brake design: know where on it your device sits. The falling magnet sits safely in the linear regime; high-speed maglev does not.

A magnet crossing a coil feels drag that peaks at v = wR/L and fades



drag opposes the relative motion

Drag rises linearly, peaks near v = wR/L, then falls: the general brake law



REMEMBER / Flux conservation sets switched-pair states; eddy drag is proportional to speed only at low speed.

Cable impedance, steps and reflections

Long cable of L prime and C prime per unit length; step excitation; matched and open ends.

The step launch

$$v = \frac{1}{\sqrt{L'C'}}, \qquad Z_0 = \sqrt{\frac{L'}{C'}}$$

A step V launched into a cable charges length C prime V v dt per time dt, drawing I = C prime V v; the same length acquires flux whose rate is L prime I v = V. Together the wave travels at 1/root(L prime C prime) and the source sees Z0 = root(L prime over C prime).

WHY / The cable looks like a resistor for as long as the front travels; energy is stored in the charged length, not dissipated.

WATCH / Source, cable and load must all read 50 ohm; mismatch any one and the scope shows a step in the trace.

$$Z_0 = 60 \ln \tfrac{b}{a}\ \Omega\ \text{(vacuum)}, \qquad v = \tfrac{c}{\sqrt{\kappa}}$$

CHECK / A load equal to Z0 absorbs the wave with no reflection because it draws exactly the delivered current.

Why 50 ohms

$$b/a = 3.5 \text{ for } 50\ \Omega \text{ in polyethylene}$$

For polyethylene with kappa = 2.3 the log gives b/a = 3.5, exactly RG-58 geometry: 2.95 mm over 0.9 mm. The 50 ohm standard compromises the maximum-power ratio about 30 ohm and the minimum-loss ratio about 77 ohm; television cable where loss matters and power does not is 75 ohm.

WHY / Standards encode physics: the numbers are not arbitrary.

WATCH / Only the impedance depends on b/a; the speed depends on the dielectric alone.

CHECK / Thirty metres of 50 ohm cable with a 10 V step: 0.2 A for the whole transit, delay 0.15 microsecond.

Reflections

$$V_{\text{reflected}} = +V \text{ (open end)}$$

Matched end: the load takes 0.2 A at 10 V and the cable carries 2 W for ever. Open end: current must vanish, so a positive reflected step doubles the voltage to 20 V behind it and the source current falls to zero on its return: the ringing of an unterminated probe.

WHY / Termination is not bookkeeping; it decides whether the delivered power is absorbed or returned.

WATCH / Short-circuit reflection doubles current instead; the two cases bracket every line problem.

$$V_{\text{open}} = 2V, \qquad I_{\text{short}} = 2I$$

CHECK / Short-circuit reflection doubles current instead. Termination decides whether delivered power is absorbed or returned: the ringing of an unterminated probe.

Fields against batteries

$$\frac{B^2}{2\mu_0} = 0.4\ \text{MJ/m}^3 \text{ at } 1\ \text{T}$$

A 1 T field stores 0.4 MJ per cubic metre; a lithium battery at 0.9 MJ/kg and 2500 kg/m^3 holds 2.2 GJ/m^3: five thousand times more. Matching batteries in a cubic metre would need 75 T and 2 GPa of containment. Fields are for power, chemistry is for energy.

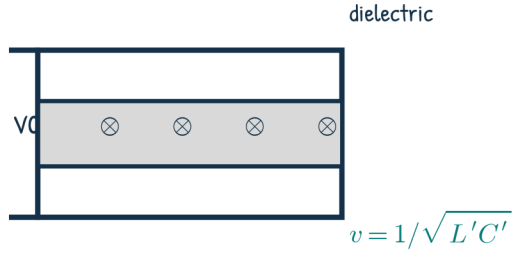
WHY / The energy-flux check: half C prime V squared plus half L prime I squared per length at speed v gives V squared / Z0, a dissipation formula for a lossless cable.

WATCH / SMES exists for power quality, megawatts for seconds, where instant response matters.

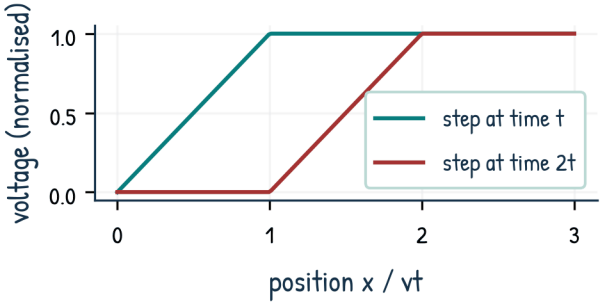
CHECK / The electric and magnetic energies of the travelling wave are equal: the cable carries power at V I with I = V/Z0.

CHECK / SMES serves power quality where response time beats density.

Coax cable: fields travel as a step at v = c/root(kappa)



A step travels at v = 1/root(L prime C prime); an open end reflects it with the same sign



REMEMBER / A cable looks like a resistance Z0 while the wave travels; 50 ohm is a compromise between power and loss.

The driven LCR: transient and steady state

Series RLC switched onto v = V0 sin omega t; uncharged capacitor; small R.

The full solution

$$i = i_s + i_h, \qquad i_s = I_0 \sin(\omega t - \phi)$$

The general solution is the phasor steady state plus a homogeneous transient A exp(-gamma t) cos(omega prime t + psi); the constants are fixed by q(0) = 0 and i(0) = 0. The transient cancels the steady state values at t = 0 and is largest when phi is large.

WHY / Transient plus steady state is the honest answer; the phasor result is only the particular solution.

WATCH / Initial conditions decide the transient amplitude, not its decay rate or frequency.

$$\gamma = \frac{R}{2L}, \qquad \omega' = \sqrt{\omega_0^2 - \gamma^2}$$

CHECK / Switch on at a voltage zero and the transient is largest; switch on at the peak and the response is nearly pure steady state.

Settling time

$$t_{1\%} \approx \frac{5 \times 2L}{R} = 25 \text{ ms}$$

For 20 ohm, 50 mH, 10 microfarad: gamma = 200/s, omega0 = 1414, omega prime = 1400 s^-1, one per cent below. The transient falls below one per cent after five time constants of 5 ms: about six cycles. A circuit with Q = 100 needs a hundred cycles to settle.

WHY / The settle time is the practical validity limit of every AC formula in the chapter.

WATCH / At exact resonance the amplitude builds monotonically over about Q/pi cycles rather than instantly.

CHECK / The natural frequency 225 Hz against the ringing 223 Hz: the transient is the free circuit, the steady state the driven.

Beats near resonance

$$|i| \text{ swells at } |\omega - \omega'|$$

Off resonance the total current is a sinusoid at omega plus a decaying one at omega prime; their sum beats while the transient lasts: the amplitude swells and shrinks a few times before settling. As omega approaches omega0 the beat slows and merges into the monotone build-up.

WHY / The beating is visible on an oscilloscope and is the fingerprint of two nearby frequencies.

WATCH / With R to zero the beat never dies; the undamped oscillator driven off resonance beats for ever.

$$R \to 0: \text{ permanent beat}; \qquad \omega = \omega_0,\ R \to 0: \text{ runaway}$$

CHECK / With R to zero the beat never dies. At exact resonance the amplitude builds monotonically over about Q over pi cycles.

The mechanical twin

$$m\ddot{x} + b\dot{x} + kx = F_0 \sin \omega t$$

A mass on a spring struck into forced oscillation from rest is this circuit symbol for symbol: transient at the natural frequency, steady state at the drive, beats in between. The analogy is exact for the linear problem and misses only radiation.

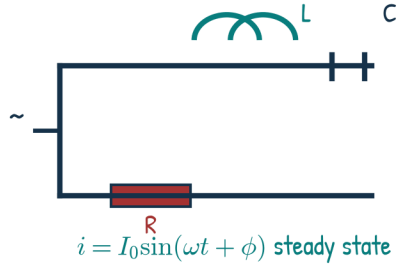
WHY / Reading the mechanics solves the circuit without new algebra: same initial-value structure.

WATCH / The energy ledger during the transient includes the kinetic term; at steady state only the average matters.

CHECK / The R to zero limit is resonance catastrophe: amplitude grows linearly without bound.

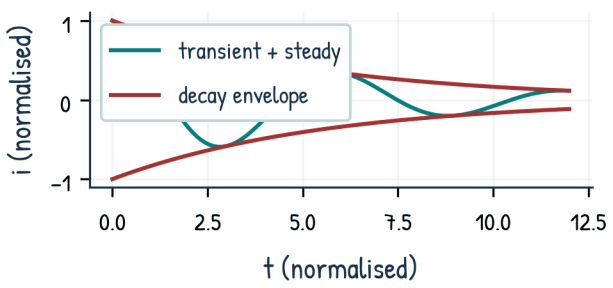
CHECK / Q counts the visible beats: Q = 20 gives about twenty oscillations inside the decay envelope.

Driven LCR: transient dies, steady state stays; watch near resonance



$i = I_0 \sin(\omega t + \phi)$ steady state

Near resonance the transient beats against the steady state before settling



REMEMBER / Every steady-state AC formula is valid only after the transient dies; near resonance the sum beats.

Maximum power transfer and three-phase

Source with internal impedance feeding a load; three equal phases 120 degrees apart.

Conjugate matching

$$Z_L = Z_s^{*}, \qquad P_{\max} = \frac{V^2}{4R_s}$$

Load power is I squared R_L with I = V over the total impedance magnitude; maximising first cancels the source reactance with the opposite kind, then matches the resistance: Z_L = R_s - j X_s. At the match half the source power is wasted in R_s.

WHY / Matching is for signal transfer, never for the grid, which runs R_L much greater than R_s for efficiency.

WATCH / The DC maximum-power theorem is the X_s = 0 special case; the AC version adds the reactance cancellation.

$$X_L = -X_s \text{ then } R_L = R_s$$

CHECK / At the match the source efficiency is exactly 50 per cent; beyond it less power arrives but a greater fraction of it.

How matching is achieved

$$Z_p = n^2 Z_s \text{ (transformer reflection)}$$

Cancel the source reactance with a power-factor capacitor for an inductive source, then transform the load resistance to R_s with a turns ratio. The loudspeaker output transformer, the aerial matching network and the 50 ohm termination are all this one move.

WHY / Impedance reflection through the transformer is the practical tool; the conjugate condition is the design goal.

WATCH / At the match, the source half-power point coincides with the delivered maximum: know the wasted half.

CHECK / An 8 ohm speaker on a 1:20 transformer looks like 3.2 kohm: matching to a transistor output stage.

CHECK / A tuner is a conjugate match that follows the aerial across the band; a fixed pad cannot do that.

Three-phase basics

$$V_{\text{line}} = \sqrt{3}\,V_{\text{phase}}$$

Three EMFs of equal amplitude 120 degrees apart sum to zero; equal loads draw currents summing to zero so the neutral carries nothing and can be thin. Line-to-line voltage is root 3 times phase voltage: 400 V between phases for 230 V phases.

WHY / The triangle of three equal phasors closes: the neutral idles under balance.

WATCH / Unbalance the loads and the neutral carries the phasor sum: never omit it in a house supply.

$$I_N = I_a + I_b + I_c = 0 \text{ (balanced)}$$

CHECK / Unbalance the loads and the neutral carries the phasor sum: never omit it in a house supply. 400 V line against 230 V phase.

Constant power

$$P(t) = 3VI\cos\phi \text{ (no ripple)}$$

Each phase delivers instantaneous power VI cos phi plus a double-frequency term; the three oscillating terms are 240 degrees apart and cancel, leaving constant total power. A three-phase motor therefore delivers steady torque where a single-phase one shudders at 100 Hz.

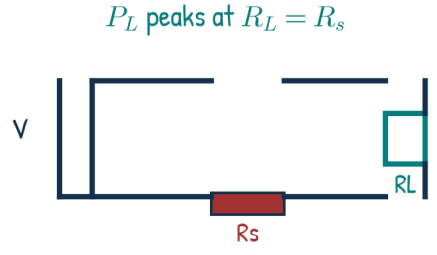
WHY / The cancellation is the industrial argument for three-phase: constant power and rotating fields.

WATCH / Single-phase motors need a start winding precisely because the pulsating power gives no defined rotation.

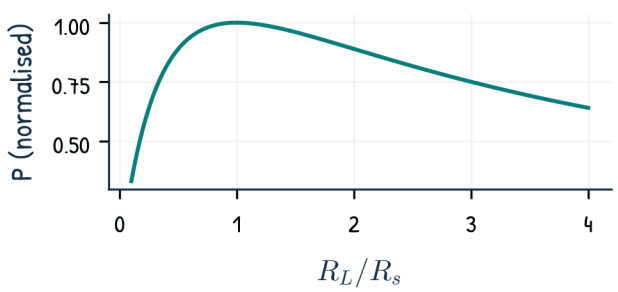
CHECK / The oscillating terms at 2 omega cancel across phases: verify with three cos squared curves 240 degrees apart.

CHECK / Three-phase power is constant in time: industrial motors run smoothly on it without any start winding.

Matched load: half the power is lost inside the source, by design



Maximum power at R_L = R_s; half the source power is wasted there



REMEMBER / Match the load impedance to the conjugate of the source; three-phase power is constant in time.

Coupled tuned circuits and the Wien bridge

Identical LC tanks coupled by mutual inductance; RC Wien network; linear amplifiers.

Normal modes

$$\omega_{\pm} = \frac{\omega_0}{\sqrt{1 \pm k}}$$

Two identical tanks coupled with M = kL have symmetric and antisymmetric modes: in-phase at omega0 / root(1+k), anti-phase at omega0 / root(1-k). For k = 0.1 the resonances sit at 0.953 and 1.054 of omega0: splitting of order k.

WHY / Coupling splits levels in circuits exactly as in coupled pendulums or quantum systems.

WATCH / k to 1 makes the anti-phase mode frequency diverge: opposing flux cancels the inductance entirely.

$$q_1 = \pm q_2 \text{ decouples the pair}$$

CHECK / For k = 0.1 the split frequencies differ by about 10 per cent. Coupling splits levels in circuits exactly as in coupled pendulums.

Critical coupling

$$k = 1/Q \text{ (flattest passband)}$$

Driven through the coupling the response is double-humped for over-coupling, single and sharp for under-coupling, and flattest with steep skirts at k = 1/Q. The intermediate-frequency transformer of a superheterodyne radio sits at exactly this point, passing one station bandwidth and rejecting neighbours.

WHY / Band-pass design is the coupling coefficient chosen against Q.

WATCH / The same physics splits the levels of two coupled pendulums; recognising the shared mathematics saves rederivation.

CHECK / For k = 0.1 the split frequencies differ by about 10 per cent: wide enough to see on a scope.

The Wien network

$$\beta = \frac{1}{3 + j(\omega RC - 1/\omega RC)}$$

A series RC feeding a parallel RC gives a transfer ratio whose imaginary part vanishes only at omega = 1/RC, where beta = 1/3 with zero phase shift. For 10 kohm and 10 nF the balance frequency is 1.6 kHz. An amplifier of gain exactly 3 with this feedback satisfies the oscillation condition at that one frequency.

WHY / The Wien-bridge oscillator is the standard low-distortion audio source; the bridge balances at one frequency and measures it.

WATCH / The network Q is only one third: purity comes from the amplifier gain control, not the network.

$$\omega_{\text{balance}} = \frac{1}{RC}, \qquad \beta = \frac{1}{3}$$

Normal coordinates

$$q_{+} = \frac{q_1 + q_2}{\sqrt{2}}, \qquad q_{-} = \frac{q_1 - q_2}{\sqrt{2}}$$

The symmetric and antisymmetric combinations decouple the loop equations directly: each is an independent oscillator at its own frequency. This is the coupled-pendulum method in circuit dress; the transfer matrix diagonalises the same way.

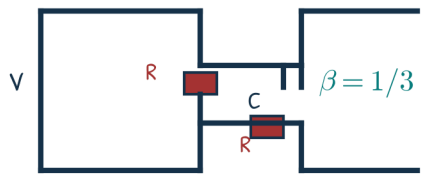
WHY / Solve in normal coordinates, transform back: the mode picture beats brute algebra.

WATCH / Check k to zero gives one frequency and k to 1 the vanishing anti-phase inductance.

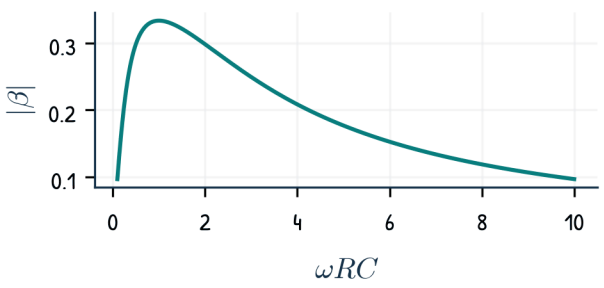
CHECK / Beta falls as one over 3 root(1 + x squared / 9) with x = omega RC - 1/omega RC: a gentle band-pass.

CHECK / The network Q is only one third: purity comes from the amplifier gain control. Beta falls gently on either side of the balance frequency.

Wien network: at omega RC = 1 the output is a third of the input, in phase



The Wien network balances at omega RC = 1 where beta = 1/3 with zero phase



REMEMBER / Two coupled resonators split their resonance; the Wien network balances at exactly one frequency.

Estimates: detectors, faults, steel and square waves

Order-unity geometry factors; material properties at working frequencies; honest error bars.

The coin and the hob

$$\delta = \sqrt{\frac{2}{\mu_0 \mu_r \sigma \omega}}$$

A 10 kHz search coil makes 31 microtesla at its centre; a coin absorbs about 7 microwatt by the lamination law with d the thickness, changing the coil resistance by 4e-4: a bridge resolves 1e-5 easily. Coupling falls as one over distance to the sixth: large coils see deep and coarse, small ones fine and shallow. The hob drives 25 kHz into iron with 0.1 mm skin and high resistance; aluminium reflects instead.

WHY / Skin depth decides cookware: ferromagnetic pans heat, aluminium mostly does not.

WATCH / The one-over-d sixth law is dipole coupling; going to distance costs more than any amplifier gains.

CHECK / Iron at 25 kHz: 0.1 mm skin; aluminium at the same frequency: 0.5 mm and much lower resistance.

The household fault

$$I_{\text{fault}} = \frac{230}{0.1} = 2.3 \text{ kA}$$

Twenty metres of house wiring has L about 20 microhenry and R about 0.1 ohm: a dead short reaches 2.3 kA with tau = L/R = 0.2 ms. Breakers need a magnetic trip opening in milliseconds as well as a thermal one, and the arc chute must swallow the half L I squared = 53 J stored in the wiring field.

WHY / The fault current is set by resistance but the timing by inductance: both trips are needed.

WATCH / The magnetic trip exists because the current is at full level within a millisecond, before any thermal element responds.

$$\tfrac{1}{2} L I^2 = 53 \text{ J in the wiring field}$$

Steel, fields and batteries

$$45 \text{ T} \rightarrow 800 \text{ MPa}$$

Magnetic pressure at 45 T is 800 MPa, the yield of high-strength steel: steady magnets stop there and pulsed machines go higher by surviving milliseconds. Field storage of 0.4 MJ/m^3 at 1 T against a battery 0.9 MJ/kg: chemistry wins a thousandfold per litre because bonds store electron-volts per atom.

WHY / Fields are for power and magnets are pressure vessels; batteries are for energy.

WATCH / SMES serves power quality where response time beats density; car storage needs chemistry.

CHECK / Holding 25 kWh in a cubic metre would need 15 T and 90 MPa containment: the estimate that kills the idea.

CHECK / A 1 T air-core store of 1 kWh fills about ten cubic metres: keep magnetic storage for milliseconds, not hours.

Square waves and model limits

$$I_{\text{rms}} = I_0 \text{ (square)}, \qquad I_{\text{rms}} = \frac{I_0}{\sqrt{3}} \text{ (triangle)}$$

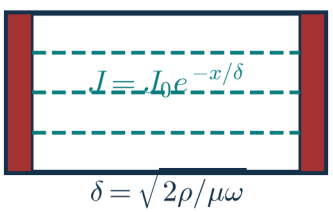
A square wave has constant i squared so its RMS equals its amplitude; the triangular wave gives I0 / root 3. A transformer will not pass a square wave unchanged: the volt-second balance forces the flux to ramp and the core saturates, distorting the tops. The model breaks for relativity, radiation, quantum spin, saturated iron and harmonics.

WHY / Compute RMS from the definition for every waveform; only the sinusoid gives 0.707.

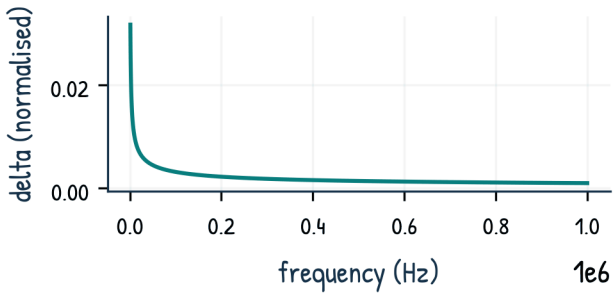
WATCH / Volt-second saturation is why transformer-coupled square waves need air gaps or reset circuits.

$$\int v \, dt = N \Delta \Phi \text{ (volt-second limit)}$$

Skin depth: current crowds into a thin shell; estimate before you compute



Skin depth falls as 1/root(frequency): iron pans heat at 25 kHz, aluminium reflects



REMEMBER / Estimates decide designs before algebra: skin depths, fault currents, containment stresses and waveform facts.